EXP_NO: 2

Sensor Node Connection Establishment

20/01/2026

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***Abstract*—Sensor node establishment is a crucial phase in wireless sensor networks (WSNs) that determines network connectivity, routing efficiency, and communication reliability. This experiment focuses on establishing a sensor network by deploying sensor nodes, forming inter-node connections, and determining an optimal communication path between selected nodes. Using graph-based modeling and Euclidean distance as a metric, the shortest path between a source and destination node is identified. The simulation highlights how node positioning and connectivity influence effective data transmission in WSNs.**

Keywords—Wireless sensor networks, sensor node establishment, connectivity, shortest path, MATLAB.

I. Introduction

Wireless sensor networks consist of multiple sensor nodes that self-organize to establish communication links for data exchange. The establishment phase includes node deployment, neighbour discovery, and path formation, which collectively impact network performance and energy efficiency.

Efficient sensor node establishment ensures reliable communication with minimal transmission cost. Graph theory is widely used to model such networks, where sensor nodes are treated as vertices and communication links are represented as weighted edges. In this experiment, sensor node establishment is demonstrated through full connectivity and shortest path identification between selected nodes, enabling efficient routing within the network.

II. EXPERIMENT

A. *Procedure*

1. Open MATLAB on the computer.
2. Create a new script file and define the network parameters.
3. Randomly deploy sensor nodes within the specified area.
4. Label each sensor node for identification.
5. Establish communication links among sensor nodes.
6. Compute distances between connected nodes.
7. Construct a weighted graph representation of the network.
8. Select source and destination nodes.
9. Identify and visualize the shortest communication path.

B. *Algorithm*

Step 1: Initialize the number of sensor nodes and deployment area.

Step 2: Generate random coordinates for sensor node placement.

Step 3: Plot and label sensor nodes in the network.

Step 4: Establish inter-node connections.

Step 5: Calculate Euclidean distance between nodes.

Step 6: Create a weighted graph model.

Step 7: Select source and destination nodes.

Step 8: Apply shortest path routing algorithm.

Step 9: Highlight the established optimal communication path.

MATLAB Program

```
clc; clear; close all;
nn = 8;
areax = 10;
areay = 10;
x = rand(1, nn) * areax;
y = rand(1, nn) * areay;
figure;
scatter(x, y, 150, 'm', 'filled');
hold on;
grid on;
for i = 1:nn
    text(x(i)+0.1, y(i)+0.1, num2str(i), ...
     'FontSize', 12, 'FontWeight', 'bold');
end
for i = 1:nn
   for j = i+1:nn
      plot([x(i) x(j)], [y(i) y(j)], 'b');
   end
end
```

```
    title('Sensor Node Connection Establishment');
    xlabel('X-axis');
    ylabel('Y-axis');
    source = 2;
    destination = 8;
    distances = Inf(nn, nn);
    for i = 1:nn
        for j = 1:nn
            if i ~= j
                distances(i,j) = sqrt((x(j)-x(i))^2 + (y(j)-y(i))^2);
            end
        end
    end
    G = graph(distances);
    [pathNodes, shortestDist] = shortestpath(G, source, destination);
    disp('Established Connection Path:');
    disp(pathNodes);
    disp(['Total Connection Distance = ', num2str(shortestDist)]);
    for k = 1:length(pathNodes)-1
        i = pathNodes(k);
        j = pathNodes(k+1);
        plot([x(i) x(j)], [y(i) y(j)], 'r', 'LineWidth', 4);
    end
hold off;
```

Result and Discussion

When Sensor nodes are randomly deployed and connected to form a communication network. The shortest communication path between source node 2 and destination node 8 is identified using graph-based routing. The established connection path minimizes transmission distance and improves routing efficiency.

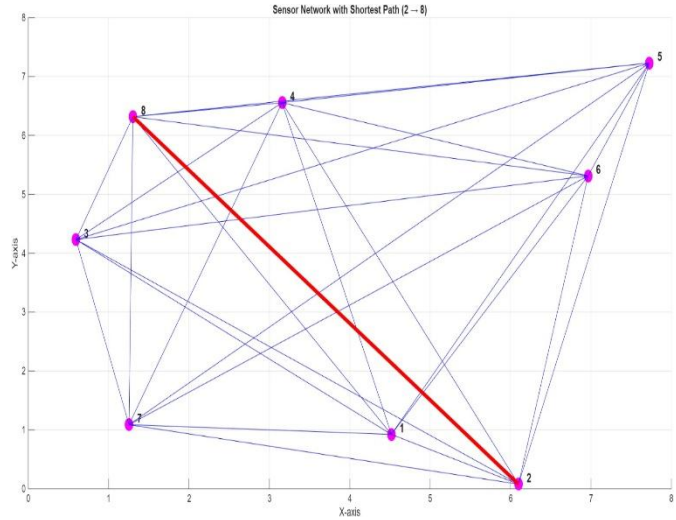

Fig. 1. Sensor node connection establishment with highlighted shortest path.

Conclusion

This experiment successfully demonstrates sensor node establishment in a wireless sensor network. By modelling the network using graph theory and identifying optimal communication paths, the importance of structured connectivity and routing efficiency is emphasized. Proper sensor node establishment enhances network reliability, reduces transmission overhead, and serves as a foundation for advanced routing and security mechanisms in WSNs.

REFERENCE

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